LECTURE 3:

THE USES OF TESTS: FUNCTIONS OF MEASUREMENT AND EVALUATION

Introduction

Welcome to lecture 3. There are so many uses of test being among them being the determination of one's progress to their next level of learning. These will be discussed in details in this particular lesson. The school-based testing that is normally a challenge when looked at from the broader perspective of applicability will also be discussed in this session. Let us review the learning outcomes first.

Learning Outcomes

After this lecture, the learner should be able to:

- Discuss the various uses of tests in educational settings.
- Explain the side effects of school-based testing

The ultimate purpose of testing is to enable the decision-making process so that improvement can be made. Testing can have many purposes. General purposes that facilitate the decision-making process that are applicable to pre-school to secondary teachers include:

1. **Placement**

Tests can be used to place students in homogenous classes or groups or streams according to their abilities. This is used to judge the capacity of students to learn and to plan for instruction that fit the learner and his capacity. That is, placement is meant for *maximum teaching efficiency*. Hence tests are used to identify exceptional children. This was the primary purpose of testing when school testing was introduced in 1905 in France (Ref. Binet IQ test- for ability grouping/streaming). – below average; average; bright

- In physical education adult fitness tests are used to determine current status so that an individualized programme can be prescribed.
- Tests are used to determine the grade or year the pupil he/she should be enrolled.

- Putting learners in either homogenous or heterogeneous groups for purposes ot instructions.

2. **Diagnosis**

Tests can be used to diagnose weaknesses or the learning difficulties of pupils while *placement* usually involves the status of the individual relative to others, diagnosis is used to isolate specific deficiencies that make for low or undesirable status. In pre-school to secondary school settings, tests can identify areas where students need to make improvements.

- For purposes of inclusive education and provision of Least Restricted Environment diagnosis with respect to hearing difficulties; visual difficulties; epilepsy and other challenges is necessary. With effect from January 2018, Grade 1 admission is accompanied by medical reports.
- In physical education / exercise or health setting, test results are used to diagnose a health problem. In PE a treadmill stress test is used to screen for heart disease. (Cite 2016 business men climbing Longonot). One died.
- In case of reading and writing, a learnerwho suffers from ***dyslexia*** writes in mirror images:

  18 for 81

  81 for 18

  F for 7

  b for d

  5 for 2

A learnerwho suffers from dyslexia speaks one thing and writes another thing.

3. **Evaluation of Achievement/Instruction**

One goal of testing is to determine whether individuals have achieved the course objectives. Placement, diagnosis and the evaluation of achievement together form the basis of individualized instruction. In pre-school to secondary school settings, this can be the achievement of instructional objectives. Testing is therefore used to assess and improve teaching.

4. **Prediction**

The test results can be used to predict the pupil's level of achievement in future activities or predict one measure from another e.g. from KCPE to KCSE performance. Prediction often seeks information on future achievement from a measure of present status, and it may help students to select the activities they are mostly likely to master.

5. **Readiness**

Pre-school tests are used to measure the child's readiness for Standard 1 tasks.

6. **Personal/Guidance and Counselling**

- The test results can be used in making decisions about the future e.g. subject choice/ career choice.
- Assist individual to make wise decision for themselves (personality tests, aptitude tests, etc.)

7. **Grading Classification**

The test results can be used to assign pupils to a particular achievement classification e.g. first class, second class and pass.

8. **Programme Evaluation**

Test results of participants can be used as one bit of evidence to evaluate the programme. By comparing the KCSE results of a County school against national norms or standards, important decisions can be made. Comparing changes in class performance between tests can provide evidence of the effectiveness of teaching.

X1 ——————————→ X2

(Test $_1$) (Test $_2$)

- In this regard, tests are used to evaluate the effectiveness of the teaching which again can lead to specific action. How effective is the teaching at Kilimo Sec. School?

Kilimo Sec KCSE Cf National KCSE Results

Results

- Used to check the progress of learning.

- How effective is the teaching at Kilimo Sec. School?

9. Motivation

Test scores can be motivating. Achievement of important standards can encourage one to achieve higher levels of performance or to participate regularly in physical activity. When children are given feedback or positive remarks on their learning progress, they are motivated.

10. Selection and Award of Scholarships

The primary purpose of national examination in developing countries including Kenya is selection- to determine [who are] the most suitable candidates for a course, a class or university. Equity Bank uses KCPE results to award scholarships to the best candidates.

- Tests are given to determine students 'ability/suitability to enter a particular school/college. What does this mean with respect to recently introduced 100% transition for KCPE candidates?

11. Evaluation of Effectiveness of Teaching Method

Which method is more effective? If the County or Country has to choose for implementation one of the methods, learners must be subjected on experimental basis to the two methods. The test results will then be used to make a decision.

Teaching Method 1 (Use of audio only)

$X_1 \longrightarrow X_2$

Pretest scores Post-test scores = Mean= 50

Teaching Method 2 (Use for both Audio and visual)

$X_1 \longrightarrow X_2$

Pretest scores Post-test scores =M=80

The question is which alternative teaching technique is efficient. From learners' scores, it appears teaching method 2 is better and hence can be adopted.

https://www.youtube.com/watch?v=VGv-AJVcuW8

SIDE EFFECTS OF TESTING

While testing has individual benefits, it has also side effects, including:

(i) The results may adversely influence the teacher's subsequent judgments. For example, even where bright student makes mistakes, the teacher can ignore.

(ii) Teachers may allow their knowledge of a child's ability in one area to affect their judgment in another area (the ***"halo effect").*** See article below that appeared in Daily Nation of Sept 13, 2014. In a country with high level of ethnicity, use of random numbers for both the candidate and the school may the remedy of "halo effect" in national examinations.

In short, "**halo effect**" is a consistent error/bias that occurs when the examiner's general impression biases his/her rating/scoring of the candidates/rates. This can be on the basis of gender, ethnicity, class ranks or knowledge of the candidate's previous performance.

The case of "Halo Effect"

No need for names on examinations papers

BY VERONICA ONJORO

Secondary school students will be sitting for their final examinations in the next one month, but most of them are disadvantaged because of the current marking system.

The system of giving candidates index numbers and then having them write their names on the examination papers results in students being disadvantaged during marking.

When a student is given index number 001, the student's total marks will never be the same as those of the student with index 030 or 080.

The reasons are because exam markers assume that 001 is the best student in that school and award them better marks, regardless of what they have written.

Abolish system

The same applies to supervisors. Whenever they want to check if the papers are marked properly, they look at the first paper, the middle paper and the last paper, assuming that the teachers who gave the index numbers gave them according to the students' abilities.

This should be abolished.

Let examiners mark papers without knowledge of the past performance of the student, so that each student has a fair chance during marking.

The other prejudice comes when students write their names on answer sheets. Some markers are known to read the names of students and determine the student's marks according to their tribes.

When the Education Bill 2014 is taken back to Parliament, let these changes be made so that all students get a fair chance during exam marking.

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Mitigation by KNEC

i. With effect from 2019, KNEC no longer require candidates to use Index Number based on mock or other school examinations but rather user the Registration Numbers given when they entered Form 1.

ii. One other way is to randomize the candidates' numbers.

(iii) ***Labeling/disabling/ stereotyping***: Labeling students as a result of testing lead to disabling them and creating stereotypes. E.g. Former KCE grading into Division 1; Division 11; Division 111; and Division IV. No matter how good DIV.IV were good in some subjects they were disabled by their label DIV. 4

(iv) ***Racism justification***. Can be used to discriminate people on the basis of race. Americans and Boers of South Africa used test results to discriminate blacks [See Arthur Jensen article attached].

(v) ***Destiny determination***. Can bring about stress on the learners and may lead to suicide in extreme case, particularly where results are of high stake in society. Eg Japan, where pupils commit suicide because of poor results. Like getting a D in KCSE in this country.

(vi) ***Political misinterpretation*** of results:

a. Kenya is notorious in accusing senior officials of KNEC when the areas where they come from did well in national examinations on the ground that they manipulated exam in favour of their home areas. The same accusation is now shifting to IEBC. Why? Everything in this country is seen from ethnic lines.

b. 1983 CPE History Paper. The following question was set in 1983 CPE History Paper:

"I was an African leader. I was overthrown by a military coup. I went to exile. I died while in exile. My remains were brought home for re-burial. Who am I?"

A. Gamal Nasser

B. Kabaka Mutesa 11

C. Haile Selassie

D. Idi Amin Dada

Though the question was well set and good in terms of being analytical and drawn from the syllabus, the question was misinterpreted by KANU politicians as having being designed in bad faith-that KNEC was inculcating overthrowing government in children's minds.

(vii). ***"self-fulfilling prophecy".*** The teacher's knowledge or expectation of a child's ability may affect the achievement of that child; the learnerwho is expected to be bright makes more progress than the learnerof whom there are lower expectations of success. This has come to be called ***"self-fulfilling prophecy".***

The teachers' expectations can create self-fulfilling prophecies. In general, self-fulfilling prophecies recur ***when false beliefs create their own reality***. In the classroom, self-fulfilling prophecy occurs when a teacher holds an initially erroneous expectation about a student, and who, through social interaction, cause the student to behave in such a manner as to confirm the originally false (but now true) expectation.

- ***In other words, this is a situation where a student becomes what the teacher expects him/her to become.***
- ***A situation where the teacher's knowledge/expectations of the learner's ability affects the future achievement of the learner.***

In the Pygmalion study, class teachers were told that the intelligence test that was administered to the children has identified 20% of the children as "late bloomers". This was not true. The 20% of the children identified to the teachers were actually selected at random.

Results after 1 year: Teacher expectations created a self-fulfilling prophecy. Behold the 20% of the children actually "bloomed" and gained more IQ points than the rest. ***Why? Possibly the teachers interacted more with those pupils and paid more attention to them in class work.***

(viii). The ***cost*** of internal testing in Kenya is passed on to parents even though it should be covered under free primary education. Even under free primary education parents are paying for these tests.

(ix). "*Leniency/severity errors*" is common in essay test. –This is where examiners/raters give ratings which that are consistently too high or too low.

(x). "*Error of central tendency".* Some examiners tend to avoid extreme categories or giving high scores, concentrating instead on categories around the midpoint or average of the scales. This is called error of central tendency. Also common in essay test.

How are some of these errors minimized?

1. Training on marking or application of the scale.
2. Adequate coordination of examiners.
3. Clear marking scheme.
4. Retiring erratic/generous/hard markers.
5. Randomizing candidates' Index Numbers
6. Blind making: Names candidates to be deleted or hidden.
7. Using school registration of candidates – do not reflect ability of learners.

   https://www.youtube.com/watch?v=rlArEdgnYKA

   https://www.youtube.com/watch?v=dPs46yOcwP0

review questions

a) With relevant examples discuss uses of tests in educational settings.

b) Explain the merits and demerits of school-based testing methods

c) Describe how tests can be used for programme evaluation

Further Reading

1. Education Measurement and Evaluation: J.Swarupa Rani, Discovery Publishing house.

2. Measurement and Evaluation in Teaching : Norman Edward Gronlund Macmillian

3. Measurement and Assessment in Teaching : Robert L. Linn Pearson Education India.

4. Program Evaluation and performance measurement : James C. Me. David, Laura R.L. Hawthorn, Sage Publications